

SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0329 – MOBILE COMPUTING

PROJECT TITLE

"Multiple Access Techniques Comparison Tool (FDMA/TDMA/CDMA/OFDMA/SDMA)"

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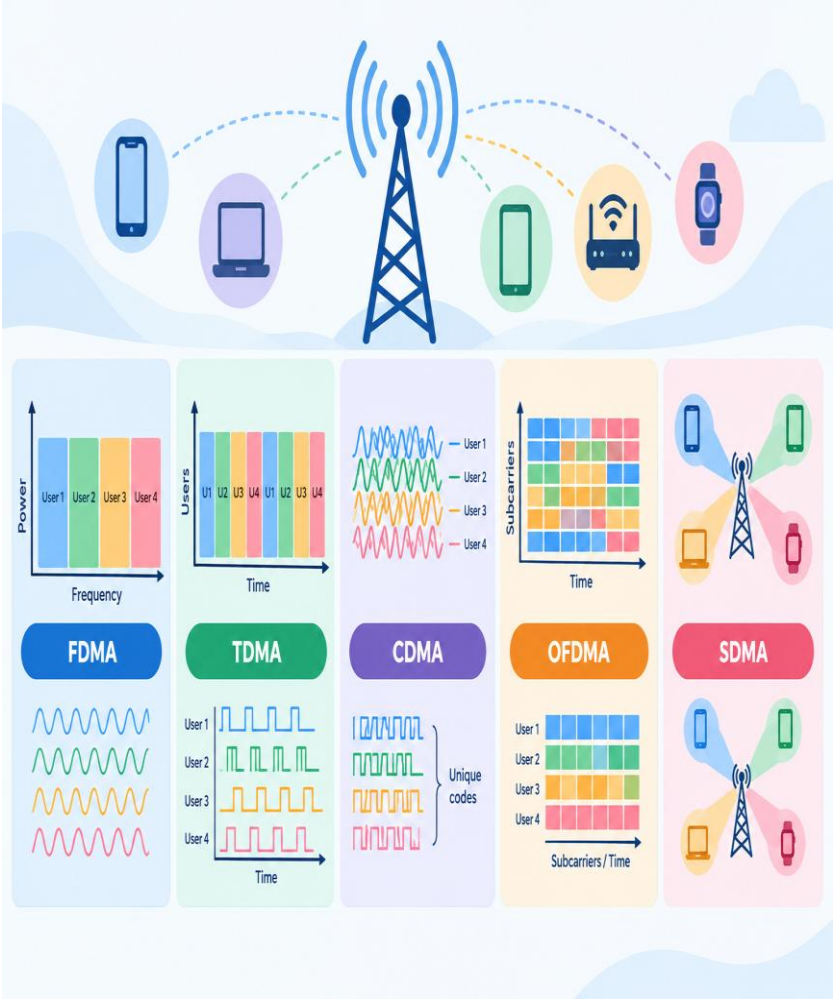
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ABSTRACT

- A Multiple Access Techniques Comparison Tool is a software tool that simulates and compares FDMA, TDMA, CDMA, OFDMA and SDMA under identical, user-defined conditions.
- The tool takes inputs such as total bandwidth, number of users, per-user data rate, SNR and channel model, and computes performance metrics for each technique.
- It computes spectral efficiency, capacity, per-user throughput, bit error rate (BER), fairness and latency, and shows them in side-by-side plots and tables.
- It helps students and engineers understand how frequency, time, code, subcarriers and space are shared among users, and the trade-offs between techniques.
- It supports faster, evidence-based selection of a suitable multiple access scheme for a given wireless scenario.

INTRODUCTION

- Multiple access techniques allow multiple users to share a wireless network.
- FDMA, TDMA, CDMA, OFDMA, and SDMA use different methods to share network resources.
- Each technique has different advantages and limitations.
- Our project simulates and compares these five techniques using different input parameters.
- The results help us understand their performance and identify suitable techniques for different wireless communication conditions.



PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

- Multiple access techniques are usually studied separately with different parameters, so direct comparison is difficult.
- The best technique depends on bandwidth, users, SNR, interference and latency; manual calculation is slow and error-prone.
- Textbook analysis is static and gives limited insight into how performance changes with parameters.

Objectives:

- To study five multiple-access techniques: FDMA, TDMA, CDMA, OFDMA, and SDMA.
- To explain the working principles of each technique.
- To compare their resource allocation methods and characteristics.
- To present their advantages, limitations, and applications.
- To develop a user-friendly tool for interactive comparison and learning.

EXISTING SYSTEM AND PROPOSED SYSTEM

Existing System:

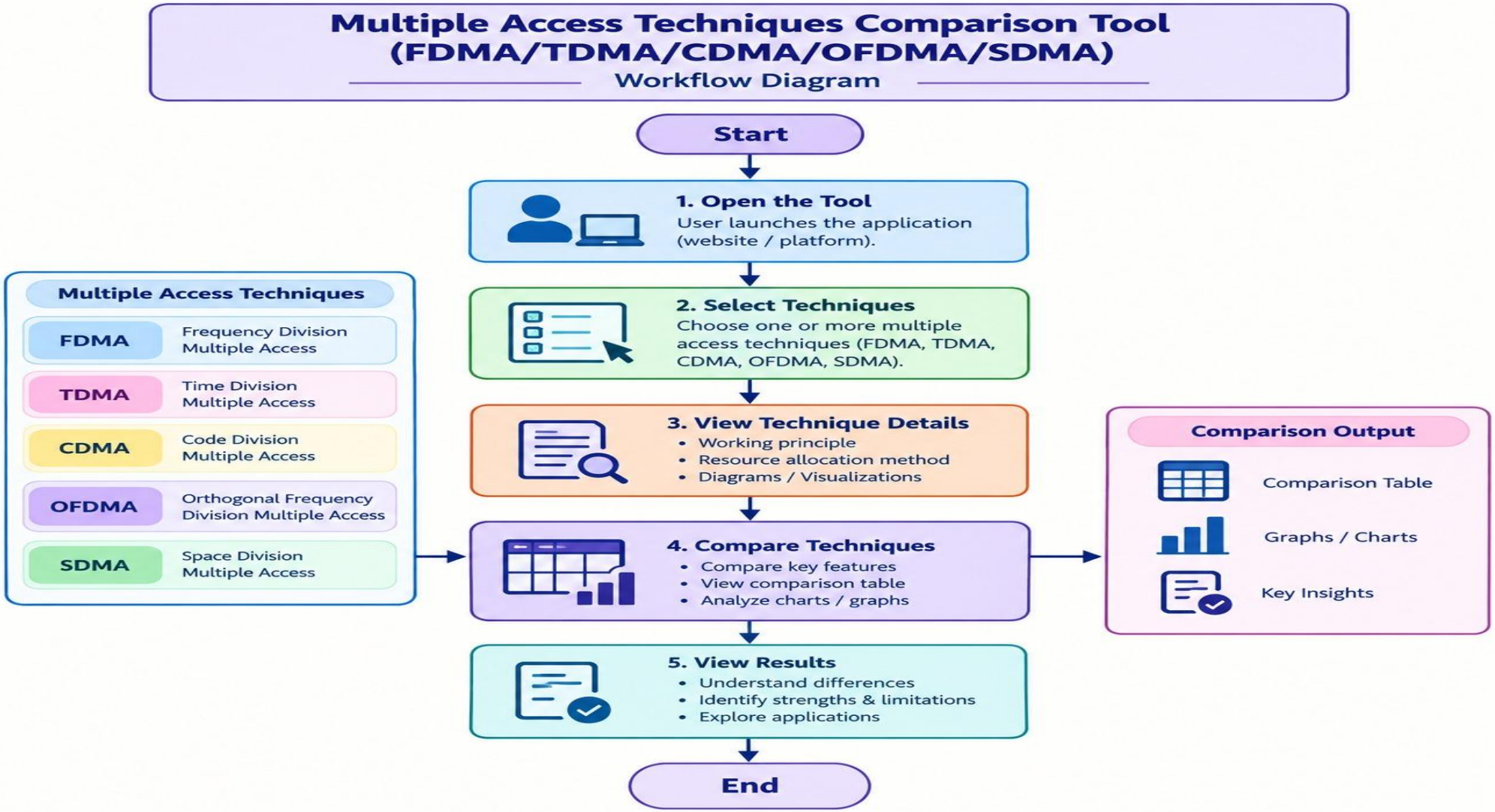
- Techniques are explained separately in textbooks and notes.
- Analysis is done by hand or with separate, one-off scripts.
- No common framework for fair comparison on the same inputs.
- Resource allocation and interference are hard to visualize.

Proposed System:

- One tool compares all five techniques under identical inputs.
- Users set bandwidth, users, SNR and channel model interactively.
- Graphs and tables show differences and trends clearly.
- Rule-based recommendation of a suitable technique.

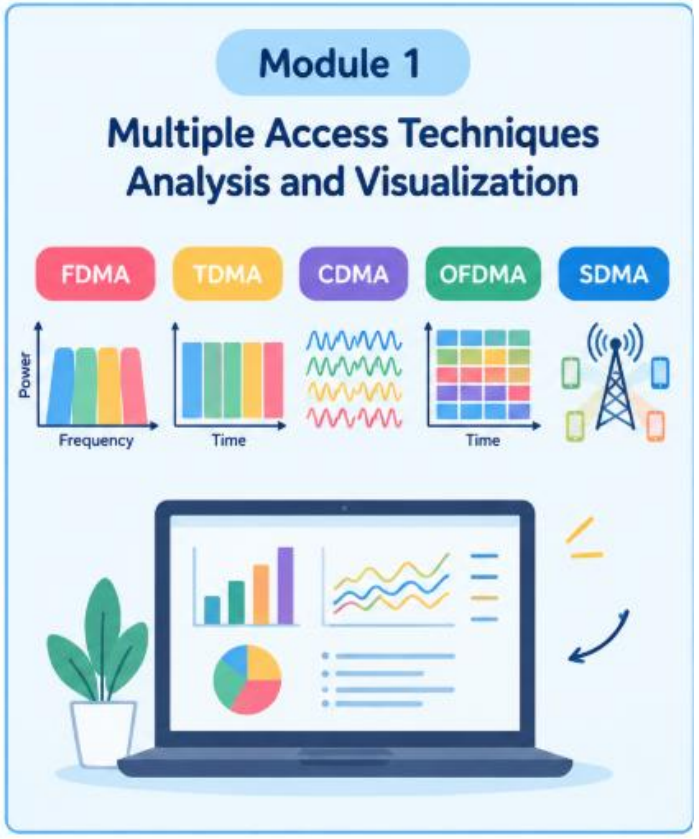
METHODOLOGY

ARCHITECTURE DIAGRAM



Module 1: Multiple Access Techniques Analysis and Visualization

- Allows users to select FDMA, TDMA, CDMA, OFDMA, and SDMA.
- Explains the working principles and resource allocation methods of each technique.
- Displays interactive diagrams to visualize how each technique works.
- Provides Play, Pause, and Reset controls for visual demonstrations.
- Explains the advantages, limitations, and applications of each technique.



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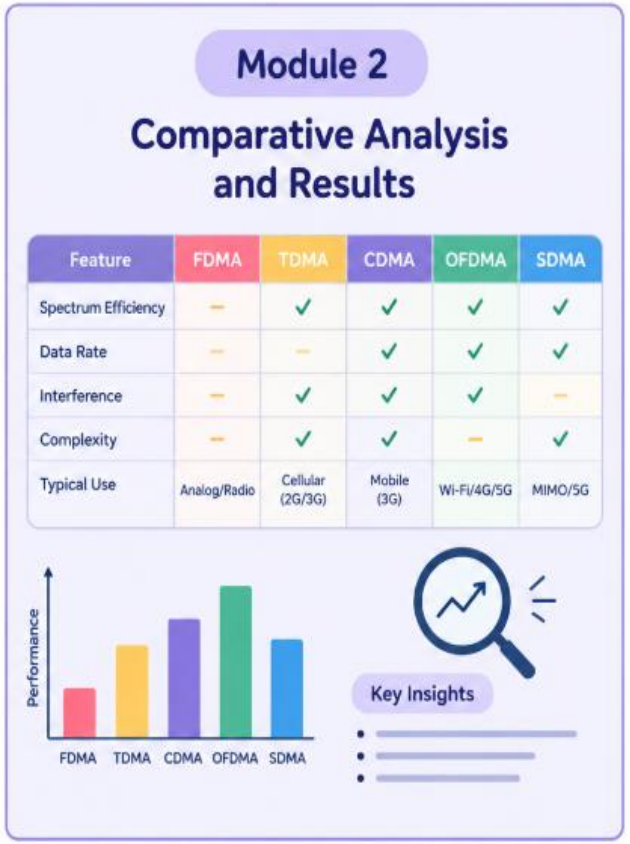
COMPARISON TABLE

Comparison of Multiple Access Techniques

Parameter	FDMA	TDMA	CDMA	OFDMA	SDMA
Full Form	Frequency Division	Time Division	Code Division	Orthogonal Frequency Division	Space Division
Separation Method	Frequency	Time	Code	Subcarriers	Space
Resource Shared	Frequency bands	Time slots	Unique codes	Frequency subcarriers	Spatial channels
Main Benefit	Simple method	Organized access	Supports multiple users	High data speed	Increases network capacity
Application	1G networks	GSM (2G)	IS-95 (2G)	4G LTE, 5G	Smart antennas, MIMO

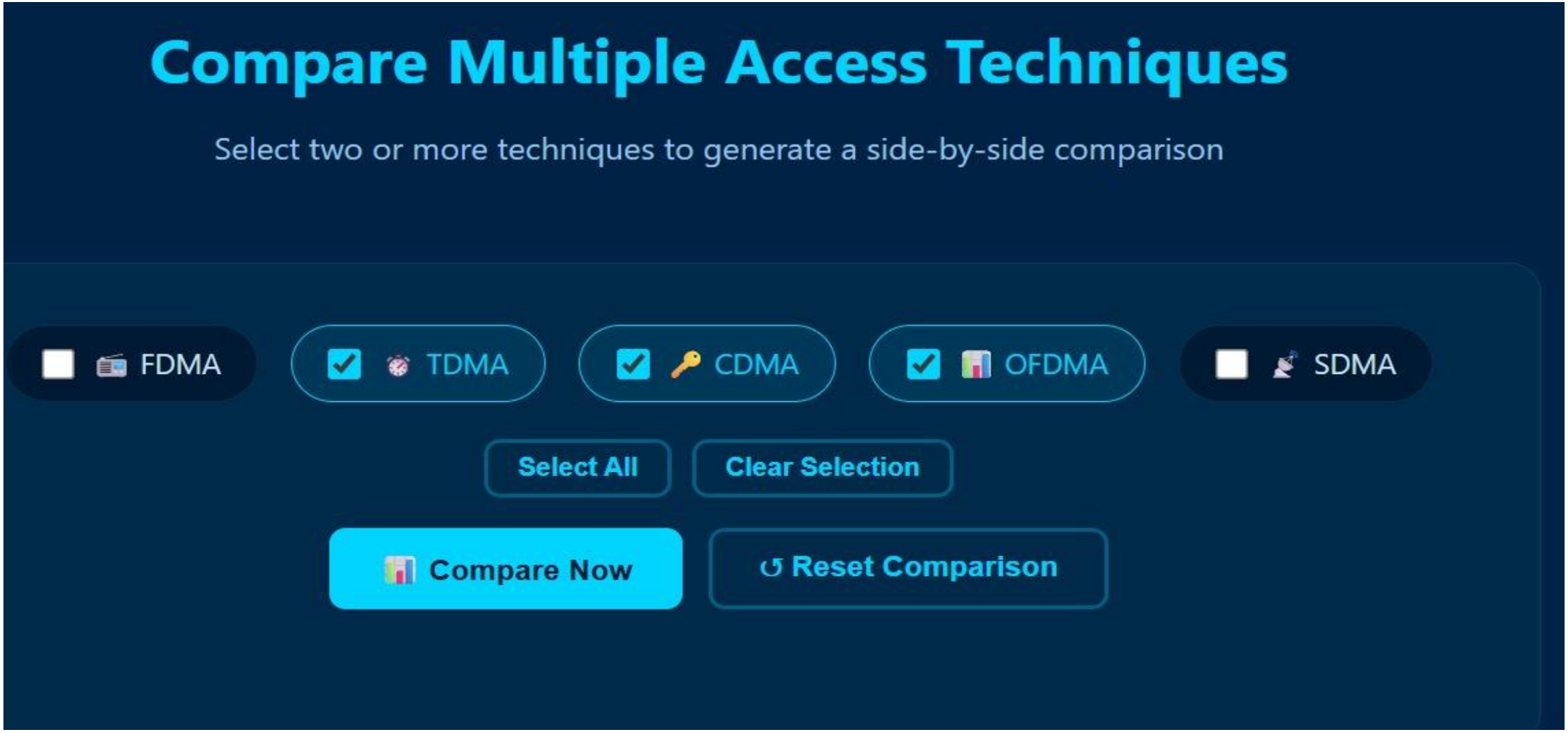
Module 2: Comparative Analysis and Results

- Allows users to select two or more techniques for comparison.
- Generates a dynamic comparison table based on the selected techniques.
- Compares resource allocation, synchronization, interference, and performance characteristics.
- Displays comparative insights highlighting the similarities and differences between techniques.
- Presents the comparison results and conclusion to help users understand the techniques.



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RESULTS



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IEEE STANDARDS

Multiple Access Standards in Practice

- **IEEE 802.11ax (Wi-Fi 6)** – Uses OFDMA to allow multiple users to transmit data simultaneously in wireless LANs.
- **IEEE 802.16 (WiMAX)** – Uses OFDMA for broadband wireless access.
- **IEEE 802.11-2020** – Defines the MAC and PHY specifications for wireless local area networks (WLANs).
- **IEEE 802.11be (Wi-Fi 7)** – Supports advanced WLAN operation, including multi-link operation and efficient resource utilization.
- **IEEE 802.16e** – Supports mobile broadband wireless access using OFDMA.
- **IEEE 802.11ac (Wi-Fi 5)** – Uses OFDM for high-speed wireless communication.
- **IEEE 802.11n (Wi-Fi 4)** – Uses OFDM and supports MIMO technology for improved WLAN performance.

SDG CONTRIBUTION

- The project supports learning and efficient use of the limited radio spectrum through clear comparison of multiple access techniques.
- **SDG 9 – Industry, Innovation, and Infrastructure**
  - Supports better design and selection of wireless communication infrastructure.
- **SDG 4 – Quality Education**
  - Gives students an interactive way to understand multiple access concepts.
- **SDG 11 – Sustainable Cities and Communities**
  - Informs efficient network design for dense urban connectivity.

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CONCLUSION

- Compares FDMA, TDMA, CDMA, OFDMA and SDMA under identical conditions.
- Quantifies capacity, spectral efficiency, BER, fairness and latency.
- Shows that no single technique is best everywhere; the choice depends on the scenario.
- Makes learning and decision-making easier through interactive visualization.
- Can be extended with realistic channel models and hybrid schemes such as NOMA [optional].
- Combines theory, simulation and visualization in one tool.

REFERENCES

1. *D. J. Deng, K. C. Chen, and R. S. Cheng, "IEEE 802.11ax: Next Generation Wireless Local Area Networks," in Proceedings of the 10th International Conference on Heterogeneous Networking for Quality, Reliability, Security and Robustness, 2014.*
EUDL

2. *Y. Daldoul, D. E. Meddour, and A. Ksentini, "Performance evaluation of OFDMA and MU-MIMO in 802.11ax networks," Computer Networks, vol. 182, article 107477, 2020.*
ScienceDirect

3. *S. Tutelian, D. Bankov, D. Shmelkin, and E. Khorov, "IEEE 802.11ax OFDMA Resource Allocation with Frequency-Selective Fading," Sensors, vol. 21, no. 18, article 6099, 2021.*
MDPI

4. *"Will OFDMA Improve the Performance of 802.11 WiFi Networks?" IEEE Wireless Communications, vol. 28, no. 3, pp. 100–107, June 2021.*
IEEE Xplore

5. *M. Aslam, X. Jiao, W. Liu, M. Mehari, T. Havinga, and I. Moerman, "A novel hardware efficient design for IEEE 802.11ax compliant OFDMA transceiver," Computer Communications, vol. 219, pp. 173–181, 2024.*

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Thank You